

Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

Computer Vision Engineer

Senior Computer Vision Engineer with 3+ years delivering production-grade vision systems in real-world, safety-critical environments. Strong ownership of end-to-end CV systems including model development, data quality frameworks, validation, and operational reliability, with close collaboration across domain experts, data engineers, and leadership. Known for taking ownership of technical decisions, setting sensible best practices, and scaling automation in a way that measurably improves accuracy and reliability.

WORK EXPERIENCE

Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Built predictive models for classifying 24/7 oil-rig operations using time-series and CV features, achieving 23% higher F1 and supporting automated risk triage under variable lighting, motion, and occlusion conditions.
- Implemented automated data ingestion, cleaning, training, and validation pipelines for large multimodal sensor datasets using MLflow, DVC, and Optuna, streamlining experimentation and reducing model iteration time by 40%.
- Developed real-time perception modules for oil-rig safety systems, deploying CNN-based vision pipelines with YOLOv5 on edge devices to improve detection accuracy by 15%.
- Designed a dynamic memory management solution to address recurring crashes in a live video streaming system, reducing crash frequency by 67%.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.
- Presented key findings and metrics to leadership and clients to translate complex performance results into actionable insights for deployment and further optimisation.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Built and optimised deep learning pipelines with YOLOv5 and DeepSORT, enabling consistent object tracking and robust analytics in dynamic industrial environments.
- Developed adversarial-condition training and validation suites simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Built and maintained large-scale image and sensor datasets, automating labelling, versioning, and performance tracking to support anomaly investigation and continuous improvement cycles.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Radiological SLAM with LiDAR Odometry

- Implemented ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Designed and implemented a real-time SLAM system integrating LiDAR and radiological data for the teleoperation of the Clearpath Husky UGV development platform.
- Built ROS2 node graph with tf2 frame management and sensor message handling to support a modular perception pipeline, enabling deterministic and repeatable evaluation via rosbag playback.
- Tuned SLAM parameters to improve localisation stability across varied terrains, achieving consistent behaviour in feature-dense indoor and feature-sparse outdoor environments.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS2 prior to real-world deployment.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 2D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Implemented ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration within a modular perception architecture.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

SKILLS

- **Perception & SLAM:** Convolutional Neural Networks, Data Augmentation, NumPy, OpenCV, SAM, Scikit-Image, U-Net, Vision Transformer (ViT), YOLO
- **Cloud, Edge & MLOps:** Apache Kafka, Docker, DVC (Data Version Control), FastAPI, Git, Linux, Microsoft Azure, MLFlow
- **Programming & Development:** Bash, C++17, Django, PostgreSQL, Python
- **Deep Learning and Optimisation:** Bayesian Optimisation, DeepSORT, MultiRocket, ONNX, OpenVINO, Optuna, PyTorch, Quantisation-Aware Training, TensorFlow, TensorRT